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An Explainable Logistic-Meta-Learner Stacking Framework with Domain-Informed Feature Engineering for Multiclass Sleep-Disorder Risk Stratification

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Abstract: Accurate risk stratification may facilitate timely referral for sleep assessment, yet models developed from small tabular datasets are particularly susceptible to outcome ambiguity, preprocessing leakage, and optimistic validation. This study develops an explainable stacking framework using the publicly available Sleep Health and Lifestyle Dataset, comprising 374 participant-level observations, 11 mixed-type predictors, and three outcome categories: no recorded disorder, insomnia, and sleep apnea. Administrative identifiers were excluded, blood pressure was decomposed into systolic and diastolic components, and every data-dependent transformation was estimated exclusively from the training partition. Random Forest and XGBoost base learners were integrated through a logistic-regression meta-learner fitted to out-of-fold class probabilities and the encoded predictor pass-through. Model behaviour was examined using complementary SHAP and LIME analyses, while discrimination was assessed on a stratified 20% hold-out set. The proposed stack correctly classified 73 of 75 observations, corresponding to an accuracy of 97.33% and a macro-F1 score of 0.96. Class-specific F1 scores were 0.93 for insomnia, 1.00 for no recorded disorder, and 0.94 for sleep apnea. Paired bootstrap analysis estimated a 1.33-percentage-point improvement over RF–XGBoost soft voting, although the confidence interval included zero and the exact McNemar test was non-significant. These findings establish a reproducible computational benchmark rather than a clinically deployable diagnostic system. Independent outcome adjudication, repeated internal validation, calibration analysis, and prospective external evaluation remain necessary.

Key words: sleep disorder, insomnia , sleep apnea, stacking ensemble, XGBoost, explainable machine learning ,clinical decision support

1. Introduction

Sleep disorders comprise clinically heterogeneous conditions with distinct diagnostic criteria and management pathways. The 2023 text revision of the International Classification of Sleep Disorders separates insomnia disorders from sleep-related breathing disorders, whereas the benchmark used here compresses this richer taxonomy into three coarse outcome categories [9]. Obstructive sleep apnea is especially relevant to screening research because conventional diagnostic pathways remain resource intensive and access to definitive testing is uneven [13, 14, 17]. A data-driven model may therefore assist referral prioritisation, but it cannot establish a diagnosis from lifestyle attributes alone.

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Structured lifestyle and physiological data offer a pragmatic basis for methodological development, but they also create substantial analytical risk. Predictors differ in scale and representation, categorical levels require harmonisation, and the outcome classes are unequal. Performance can be materially inflated when preprocessing precedes data partitioning, model choices are adapted to evaluation results, or uncertainty from a small test sample is ignored [12, 23, 26]. Reliable evaluation consequently requires strict separation of development and test data, class-sensitive metrics, paired uncertainty analysis, and explicit limits on transportability and clinical interpretation.

This study addresses these requirements through five contributions. First, it specifies an auditable preprocessing strategy for mixed numerical and categorical variables, with all learned transformations confined to the development partition. Second, it develops a heterogeneous stack in which Random Forest, XGBoost, and a radial-basis-function support vector machine contribute out-of-fold class probabilities and the encoded predictors are passed to a logistic meta-learner. Third, it evaluates the proposed model against soft voting and previously reported ANN ensembles, including a repeated-split robustness check. Fourth, it combines SHAP, LIME, a PCA-based class-separability diagnostic, and a two-dimensional Accumulated Local Effects surface to examine global importance, local attribution, and feature-interaction patterns while explicitly separating predictive association from causal or diagnostic interpretation [20, 21, 30]. Fifth, it reports one-vs-rest ROC–AUC alongside threshold-based metrics to characterise ranking quality independently of the classification threshold.

2. Related work and study rationale

Recent sleep-AI research has concentrated primarily on apnea detection from physiological signals and wearable devices. Contemporary reviews cover electrocardiographic, oximetric, respiratory, and multimodal signal pipelines, and consistently distinguish automated screening from clinically adjudicated diagnosis [13–16]. This evidence base is informative for clinical framing but is not directly comparable with the present low-dimensional lifestyle dataset, which contains neither raw physiological waveforms nor polysomnographic severity indices.

For tabular prediction, recent evidence continues to treat tree ensembles and neural architectures as complementary rather than universally interchangeable model families [18]. Alom et al. applied ANN bagging, boosting, and weighted averaging to the same sleep-health benchmark used here [10]; their reported accuracies provide the only literature-derived numerical comparator in this manuscript. In a different clinical setting, Alabi et al. demonstrated that heterogeneous stacking can be paired with SHAP and LIME, but their cancer-survival task, sample construction, and validation design preclude direct performance comparison [19]. The unresolved methodological question is therefore whether an out-of-fold RF–XGBoost–SVM stack can improve discrimination on this specific benchmark while maintaining strict evaluation separation and transparent uncertainty.

The explanation layer is deliberately bounded. Recent reviews emphasise that post-hoc explanations require explicit definitions of audience, fidelity, stability, and intended use; an attribution is evidence about a fitted prediction function, not a causal mechanism or clinical rationale [20, 21]. Likewise, current reporting and deployment guidance requires transparent data provenance, evaluation design, uncertainty, and validation before clinical use [27, 28]. Accordingly, all ANN percentages in the comparative table are transcribed from Alom et al. [10], whereas every RF–XGBoost–SVM value is generated by the present analysis. The comparison is descriptive because the studies do not share an identical evaluation protocol.

Table 1. Summary of related studies and their relevance to this work.

Study	Data/task	Methods	Relevance
Alom et al. [10]	Sleep-health tabular data	ANN Ensembles	Main Numerical Comparator
Alabi et al. [19]	Cancer survival	Stacking, SHAP, LIME	Supports explainable stacking
Bazoukis et al. [14]	Physiological sleep data	ML/DL methods	Clinical screening context
Tyagi et al. [16]	Sleep signals	ML/DL pipelines	Signal-based comparison
Borisov et al. [18]	Tabular datasets	Tree and neural models	Supports heterogeneous models
This study	**Sleep-Health Tabular Data**	**RF–XGBoost–SVM stack**	**TVLE + XAI + Uncertainty**

3. Materials and Methods

3.1. Methodological overview

The investigation followed a leakage-controlled analytical pathway (Fig. 1). Data integrity and outcome coding were examined before modelling, after which the administrative identifier was removed and clinically interpretable features were derived. A stratified test partition was then isolated prior to fitting any transformation. Within the development partition, numerical variables were standardised, categorical variables were one-hot encoded, and out-of-fold probability estimates were generated for both base learners. These estimates and the encoded predictors formed the input to a logistic-regression meta-learner through the implemented pass-through connection. Final discrimination, paired model comparisons, and explanation analyses were conducted only after the complete pipeline had been fixed. This design preserves the independence of the test set and prevents information from the evaluation sample from influencing representation learning or model construction.

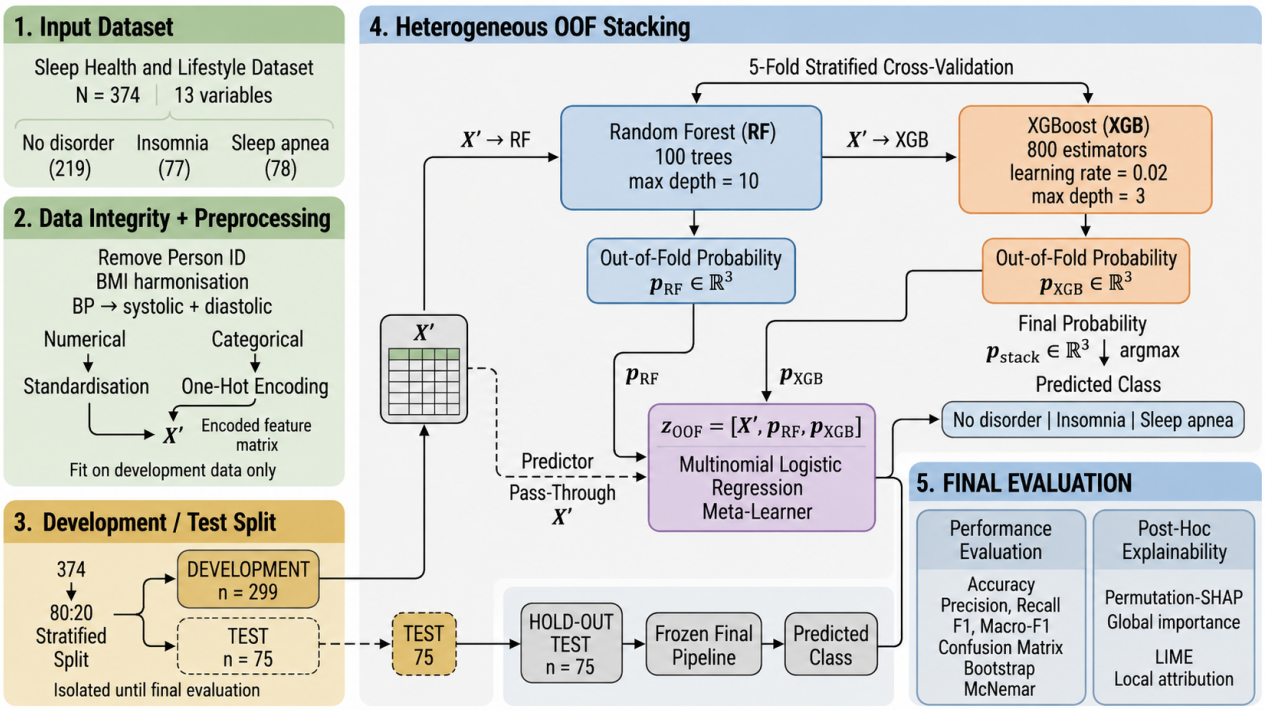


Figure 1. End-to-end framework for leakage-controlled, explainable RF–XGBoost stacking. Stage 1–2 correspond to Section 3.3 (data source and outcome definition) and Algorithm 1; Stage 3 corresponds to the development–test partition of Eq.(2); Stage 4 corresponds to the out-of-fold stacking architecture of Section 3.5 and Algorithm2; Stage 5 corresponds to the evaluation protocol of Section 4 and the explainability procedures of Section 3.6.

3.2. Problem formulation and notation

Let the benchmark sample be $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, where $N = 374$, $\mathbf{x}_i \in \mathbb{R}^{d_n} \times \mathcal{A}^{d_c}$ contains numerical and categorical predictors, and $y_i \in \mathcal{C} = \{0, 1, 2\}$ denotes insomnia, no recorded disorder, or sleep apnea. The analytical objective is to estimate a probabilistic mapping

$$F : \mathcal{X} \to \Delta^{C-1}, \qquad \hat{y}_i = \arg\max_{c \in \mathcal{C}} F_c(\mathbf{x}_i), \tag{1}$$

where $C = 3$ and Δ^{C-1} is the probability simplex. The stratified partition satisfies

$$\mathcal{D}_{tr} \cap \mathcal{D}_{te} = \varnothing, \qquad |\mathcal{D}_{tr}| = 299, \qquad |\mathcal{D}_{te}| = 75, \tag{2}$$

and every estimated parameter is a function of $\mathcal{D}_{tr}$ only. For numerical predictor j, development-set standardisation is

$$\widetilde{x}_{ij} = \frac{x_{ij} - \mu_j^{(tr)}}{\sigma_j^{(tr)}}, \quad \mu_j^{(tr)} = \frac{1}{n_{tr}} \sum_{i \in \mathcal{D}_{tr}} x_{ij}, \tag{3}$$

where $\sigma_j^{(tr)}$ is the sample standard deviation estimated from $\mathcal{D}_{tr}$. A categorical level a_q is represented by $o_{iq} = \mathbb{I}(x_{ij} = a_q)$. Consequently, the fitted transformer $T_{\widehat{\theta}_{tr}}$ maps each observation to the fixed-schema vector $\mathbf{x}'_i = T_{\widehat{\theta}_{tr}}(\mathbf{x}_i)$, with neither the scale parameters nor the categorical vocabulary re-estimated on the test partition.

3.3. Data source and outcome definition

The analysis used the publicly available Sleep Health and Lifestyle Dataset hosted on Kaggle [11]. The dataset comprises 374 participant-level observations and 13 variables: one administrative identifier, 11 candidate predictors, and the Sleep Disorder outcome. The predictors represent demographic characteristics, sleep behaviour, physical activity, perceived stress, body-mass category, heart rate, and blood pressure. Because the data were not collected as a prospectively recruited clinical cohort and the outcome was not independently adjudicated, the dataset is treated as a methodological benchmark for risk stratification rather than as a source of prevalence estimates or diagnostic ground truth.

All 11 predictors were complete, whereas 219 outcome entries were blank. Consistent with the coding used in the analytical dataset, blank outcomes were assigned to the no-recorded-disorder category, producing 219 observations with no recorded disorder, 77 with insomnia, and 78 with sleep apnea. This operational definition is consequential because absence of a recorded label is not equivalent to clinician-confirmed absence of disease. The assignment was therefore retained for reproducibility but treated as a primary source of outcome uncertainty throughout interpretation.

Table 2. Descriptive characteristics of the 374 observations used for model development and evaluation.

Profile item	Observed value	Interpretation
Records and variables	374 records; 13 variables	Person-level tabular benchmark
Candidate predictors	11	Mixed numerical and categorical fields
Outcome completeness	155 labelled; 219 blank	Blank entries mapped to no recorded disorder
Class distribution	No recorded disorder, 219; insomnia, 77; sleep apnea, 78	Unequal class frequencies
Age	27–59 years; mean, 42.18	Demographic predictor
Sleep duration	5.8–8.5 h; mean, 7.13 h	Self-reported sleep behaviour
Heart rate	65–86 bpm; mean, 70.17 bpm	Physiological indicator
Daily steps	3,000–10,000; mean, 6,816.84	Activity indicator
Categorical levels	Gender, 2; occupation, 11; BMI, 4	Harmonised before one-hot encoding

Age ranged from 27 to 59 years (mean 42.18, standard deviation 8.67), and reported sleep duration ranged from 5.8 to 8.5 h (mean 7.13, standard deviation 0.80). Quality-of-sleep scores ranged from 4 to 9, physical-activity levels from 30 to 90, stress scores from 3 to 8, heart rate from 65 to 86 beats/min, and daily step counts from 3,000 to 10,000. The categorical variables comprised two gender categories, 11 occupational groups, and four body-mass-index labels. Blood pressure, originally represented as a systolic/diastolic string, was separated into two numerical variables. The synonymous labels “Normal Weight” and “Normal” were consolidated before one-hot encoding.

Person ID was excluded because it contains ordering information without a plausible physiological interpretation. Occupation was retained as a contextual predictor, although its contribution requires careful scrutiny because it may encode socioeconomic structure and introduce transportability concerns. The public-source provenance, modest sample size, outcome-assignment rule, and absence of clinical adjudication therefore define the inferential boundaries of this study.

Table 3. Predictor groups, representative variables, and their computational treatment within the modelling pipeline.

Variable group	Examples	Treatment
Administrative	Person ID	Removed before modelling
Demographic	Age, Gender, Occupation	Categorical encoding; age standardisation
Sleep behaviour	Sleep Duration, Quality of Sleep	Numerical standardisation
Lifestyle	Physical Activity Level, Daily Steps	Numerical standardisation
Psychophysiological	Stress Level, Heart Rate	Numerical standardisation
Anthropometric	BMI Category	Category harmonisation and one-hot encoding
Cardiovascular	Blood Pressure	Split into systolic and diastolic values
Target	Sleep Disorder	Label encoding into three classes

3.4. Class separability diagnostic (PCA)

To characterise the intrinsic separability of the three outcome classes independent of any specific classifier, a two-component principal component analysis was fitted to the fully preprocessed (scaled and one-hot encoded) feature matrix using all 374 observations. This projection is diagnostic only: it is fitted separately from, and never substituted into, the classification pipeline described in Sections 3.5–3.6, so it does not affect any reported classification metric.

The first two components explained 33.68% and 29.47% of total variance respectively (63.14% combined), indicating that a large share of the feature-space structure is not captured in two dimensions and that any apparent overlap in the projection understates the information available to the full-dimensional classifiers. Fig. 2a shows the raw two-dimensional scatter, and Fig. 2b overlays 1-sigma covariance ellipses per class. The no-recorded-disorder class occupies a visually distinct region, consistent with its perfect recall in Table 5, whereas the insomnia and sleep-apnea ellipses show partial overlap. This is consistent with, though not a direct explanation for, the two insomnia-to-apnea misclassifications reported in Section 5 and discussed in Section 7.

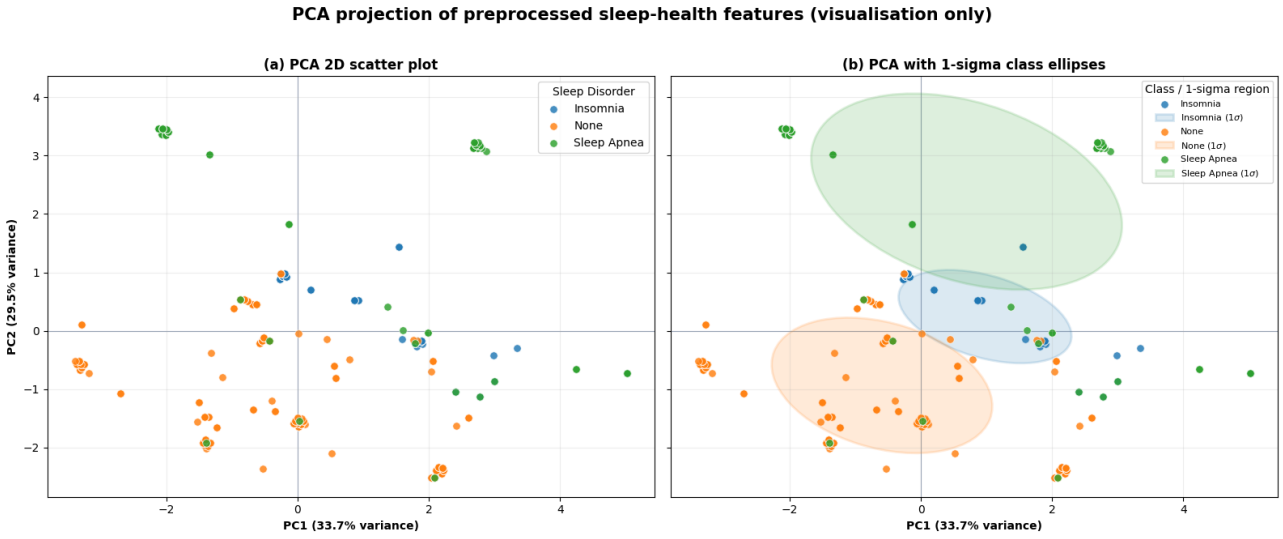


Figure 2. PCA projection of the preprocessed feature space (diagnostic only, not part of the modelling pipeline): (a) raw scatter by outcome class; (b) the same projection with 1-sigma class ellipses. PC1 and PC2 explain 33.68% and 29.47% of variance, respectively.

3.5. Data splitting and leakage control

An 80:20 stratified partition with random seed 42 produced a development set of 299 observations and an independent test set of 75 observations. The split was performed before numerical scaling, categorical encoding, or model fitting. All preprocessing parameters that includes numerical means and variances, categorical vocabularies, and label mappings. These were estimated from the development data and subsequently applied without modification to the test set. The same principle was maintained within cross-validation folds used to construct the stack. This ordering prevents evaluation-set information from influencing either the feature representation or the fitted decision function [12, 26].

Algorithm 1 Leakage-Controlled Data Preparation and Feature Transformation

Require: Dataset D, target y, random seed $s = 42$
Ensure: Transformed development and test partitions (X'_{tr}, y_{tr}) and (X'_{te}, y_{te})

1: Remove administrative identifier fields from D
2: Assign blank outcome entries to the "no recorded disorder" class
3: Harmonise categorical labels, including BMI categories
4: Decompose Blood Pressure into systolic and diastolic numerical features
5: Perform an 80:20 stratified split:

$$(D_{tr}, D_{te}) \leftarrow \text{StratifiedSplit}(D, y, s)$$

6: Extract:

$$(X_{tr}, y_{tr}),\ (X_{te}, y_{te})$$

7: Fit numerical standardisation parameters (μ_{tr}, σ_{tr}) using X_{tr} only
8: Fit the categorical one-hot encoder $\mathcal{E}_{tr}$ using X_{tr} only
9: Construct the training transformation:

$$X'_{tr} \leftarrow \mathcal{E}_{tr}\left(\frac{X_{tr} - \mu_{tr}}{\sigma_{tr}}\right)$$

10: Apply the same fitted transformations to the held-out data:

$$X'_{te} \leftarrow \mathcal{T}_{tr}(X_{te})$$

11: Fit the label mapping $\mathcal{L}_{tr}$ using y_{tr} only
12: Encode:

$$y_{tr} \leftarrow \mathcal{L}_{tr}(y_{tr}), \qquad y_{te} \leftarrow \mathcal{L}_{tr}(y_{te})$$

13: Preserve the resulting feature schema for both partitions
14: **return** (X'_{tr}, y_{tr}) and (X'_{te}, y_{te})

3.6. Base Learners and Stacking Design

The proposed architecture combines two tree-based learners with complementary bias–variance characteristics. The Random Forest comprised 100 trees with a maximum depth of 10, a minimum split size of 5, the default minimum leaf size of 1, and random seed 42. For B fitted trees, its class probability is the average tree vote

$$p_{RF}(c \mid \mathbf{x}') = \frac{1}{B}\sum_{b=1}^{B} \mathbb{I}(h_b(\mathbf{x}') = c)\,. \tag{4}$$

XGBoost used 800 boosting iterations, a learning rate of 0.02, maximum depth 3, row and column subsampling rates of 0.90, minimum child weight 3, $\gamma = 0.15$, and $\lambda = 1.2$. Its class-specific additive score and corresponding probability can be written as

$$s_c^{(M)}(\mathbf{x}') = \sum_{m=1}^{M} \eta f_{mc}(\mathbf{x}'), \qquad p_{XGB}(c \mid \mathbf{x}') = \frac{\exp(s_c^{(M)})}{\sum_{r=1}^{C}\exp(s_r^{(M)})}. \tag{5}$$

For each observation in the development set, five-fold cross-validation generated probabilities from models that excluded that observation during fitting. If $\kappa(i)$ denotes the fold containing observation i, the meta-level vector is

$$\mathbf{z}_i^{OOF} = \left[\mathbf{x}_i'^{\top},\ \mathbf{p}_{RF}^{(-\kappa(i))}(\mathbf{x}_i')^{\top},\ \mathbf{p}_{XGB}^{(-\kappa(i))}(\mathbf{x}_i')^{\top} \right]^{\top} \in \mathbb{R}^{d+2C}. \tag{6}$$

The inclusion of $\mathbf{x}_i'$ represents the implemented pass-through connection. A multinomial logistic meta-learner then estimates

$$p_S(c \mid \mathbf{z}) = \frac{\exp(\beta_{c0} + \boldsymbol{\beta}_c^{\top}\mathbf{z})}{\sum_{r=1}^{C} \exp(\beta_{r0} + \boldsymbol{\beta}_r^{\top}\mathbf{z})}, \tag{7}$$

by minimising regularised multiclass log loss

$$\widehat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left[-\frac{1}{n_{tr}} \sum_{i=1}^{n_{tr}} \sum_{c=1}^{C} \mathbb{I}(y_i = c) \log p_S(c \mid \mathbf{z}_i^{OOF}) + \lambda_\beta \sum_{c=1}^{C} \|\boldsymbol{\beta}_c\|_2^2 \right]. \tag{8}$$

Here, λ_β denotes the L2 penalty governed by the implementation's inverse-regularisation setting $C_{\mathrm{LR}} = 2.0$. The base models were subsequently refitted on the full development set for inference on unseen observations. Restricting meta-learner training to out-of-fold predictions is essential; using in-sample probabilities would expose the second level to optimistically biased representations [12, 19].

Table 4. Hyperparameter configuration of the Random Forest and XGBoost base learners, logistic-regression meta-learner, and validation split.

Component	Parameter	Value
Random Forest	number of trees	100
	maximum depth	10
	minimum split / leaf	5 / 1
XGBoost	estimators / learning rate	800 / 0.02
	maximum depth	3
	row / column subsample	0.90 / 0.90
	min child weight / γ	3 / 0.15
Meta-learner	model	logistic regression
	inverse regularisation / solver	$C_{\mathrm{LR}} = 2.0$ / L-BFGS
	encoded-feature pass-through	enabled
Validation	split / seed	80:20 stratified / 42
	stacking folds	5

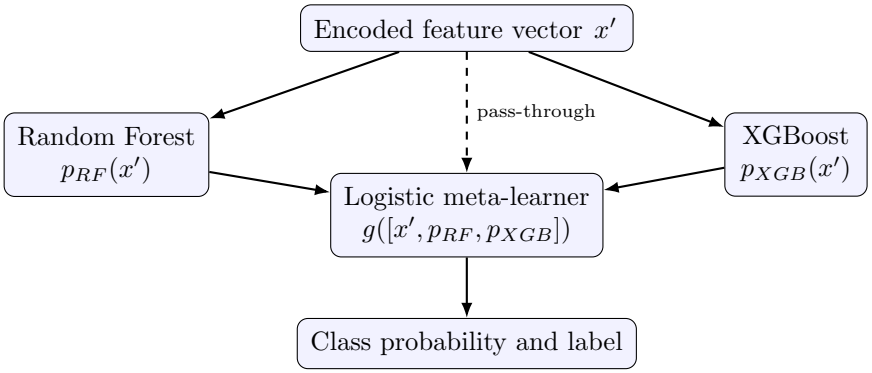


Figure 3. Architecture of the proposed heterogeneous stack. Out-of-fold Random Forest and XGBoost class-probability vectors are concatenated with the encoded predictors through the pass-through connection; the logistic meta-learner then estimates the final multiclass probabilities.

Algorithm 2 Out-of-Fold RF–XGBoost Stacking

Require: Development data (X'_{tr}, y_{tr}), test vector x', number of folds $K = 5$
Ensure: Final predicted class $\hat{y}$

Partition (X'_{tr}, y_{tr}) into K stratified folds
Initialize out-of-fold probability matrices $P_{RF}, P_{XGB} \in \mathbb{R}^{n_{tr} \times C}$
for $k = 1$ to K **do**
    Let $\mathcal{D}^{(k)}_{train} = \mathcal{D}_{tr} \setminus \mathcal{D}^{(k)}$ and $\mathcal{D}^{(k)}$ be the validation fold
    Train Random Forest $f^{(-k)}_{RF}$ on $\mathcal{D}^{(k)}_{train}$
    Train XGBoost $f^{(-k)}_{XGB}$ on $\mathcal{D}^{(k)}_{train}$
    Obtain validation probabilities:

$$P^{(k)}_{RF} = f^{(-k)}_{RF}(X'^{(k)}), \qquad P^{(k)}_{XGB} = f^{(-k)}_{XGB}(X'^{(k)})$$

    Store $P^{(k)}_{RF}$ and $P^{(k)}_{XGB}$ in their corresponding out-of-fold positions
end for
Construct the meta-level representation:

$$Z = \left[X'_{tr}, P_{RF}, P_{XGB}\right]$$

Train multinomial logistic-regression meta-learner $g(Z, y_{tr})$ with $C_{\mathrm{LR}} = 2.0$
Refit RF and XGBoost on the complete development set:

$$f_{RF} \leftarrow f_{RF}(X'_{tr}, y_{tr}), \qquad f_{XGB} \leftarrow f_{XGB}(X'_{tr}, y_{tr})$$

Generate test-set base-learner probabilities:

$$p_{RF} = f_{RF}(x'), \qquad p_{XGB} = f_{XGB}(x')$$

Construct the test meta-vector:

$$z^* = [x', p_{RF}, p_{XGB}]$$

Compute final class probabilities:

$$p^* = g(z^*)$$

Select the class with maximum posterior probability:

$$\hat{y} = \arg \max_{c \in \{1,\ldots,C\}} p^*_c$$

return $\hat{y}$

3.7. Explainability and Auditability

SHAP explanations were obtained through a permutation-based, model-agnostic interface. A fixed background sample B was drawn exclusively from the development partition, and each perturbed observation was passed through the complete preprocessing and prediction pipeline. For class c, local additive consistency satisfies

$$F_c(\mathbf{x}_i) - \mathbb{E}_{\mathbf{X}\sim B}[F_c(\mathbf{X})] = \sum_{j=1}^{d} \phi_{ij}^{(c)}, \tag{9}$$

and global importance for feature j was defined as

$$I_j = \frac{1}{nC} \sum_{i=1}^{n} \sum_{c=1}^{C} |\phi_{ij}^{(c)}|. \tag{10}$$

Here, $\phi_{ij}^{(c)}$ denotes the attribution assigned to feature j for observation i and class c. LIME was used as a complementary local analysis. Given perturbed neighbourhood $\mathcal{Z}_i$, proximity kernel π_i, and interpretable surrogate family $\mathcal{G}$, the local explanation is

$$g_i^* = \arg\min_{g\in\mathcal{G}} \left\{ \sum_{\mathbf{z}\in\mathcal{Z}_i} \pi_i(\mathbf{z})[F_c(\mathbf{z}) - g(\mathbf{z})]^2 + \Omega(g) \right\}, \tag{11}$$

where $\Omega(g)$ penalises surrogate complexity. Absolute LIME coefficients were aggregated only to assess broad agreement with SHAP rankings; they were not treated as a second estimate of causal importance. Both methods characterise the fitted prediction function, and their outputs depend on the chosen background distribution, perturbation mechanism, and representation of predictor dependence [20, 22].

Algorithm 3 Permutation-SHAP Procedure for Global Model Auditing

Require: Fitted prediction pipeline F, training matrix X_{tr}, explanation sample X_e
Ensure: Ranked global feature-importance vector I

1: Sample a fixed background set:
$$B \subset X_{tr}$$
2: Define the complete prediction function:
$$h(z) = F(\text{decode}(z))$$
3: Initialise the permutation-based SHAP explainer using h and background B
4: Evaluate SHAP values for all observations in X_e:
$$\Phi = \text{SHAP}(h, X_e, B)$$
5: **for** each feature $j = 1, \ldots, d$ **do**
6: Aggregate absolute class-specific contributions:
$$I_j \leftarrow \frac{1}{n_e C} \sum_{i=1}^{n_e} \sum_{c=1}^{C} \left|\Phi_{ij}^{(c)}\right|$$
7: **end for**
8: Rank features in descending order of I_j
9: **return** Ranked importance vector I

3.8. Feature relationships and stacking-model local effects

Two additional diagnostics were used to examine dependence structure among predictors and the fitted stack's local behaviour. First, Spearman rank correlation was computed across the nine numerical predictors on the development partition, to characterise monotonic association independent of the standardisation applied in Eq. (3). Second, a two-dimensional Accumulated Local Effects (ALE) surface was computed directly from the fitted stacking pipeline for the joint effect of Sleep Duration and Quality of Sleep on the predicted probability of insomnia, using a 6×6 quantile grid. Unlike SHAP, ALE conditions on the local data density and is therefore less sensitive to feature correlation when estimating joint effects; it was accumulated by summing corner-difference finite effects across grid cells and centring the resulting surface to zero mean.

Fig. 4a shows that Sleep Duration and Quality of Sleep are strongly positively correlated, while Stress Level is negatively correlated with both, consistent with their combined appearance among the higher-ranked SHAP features in Section 3.7 and Section 5.3. Fig. 4b shows the corresponding ALE surface: the predicted insomnia probability rises most sharply in the region of short sleep duration combined with low sleep quality, and flattens once both variables exceed their sample medians, indicating a joint rather than additive relationship between these two predictors and the insomnia class score. As with SHAP and LIME, the ALE surface characterises the fitted function's local behaviour and should not be read as a causal dose-response relationship.

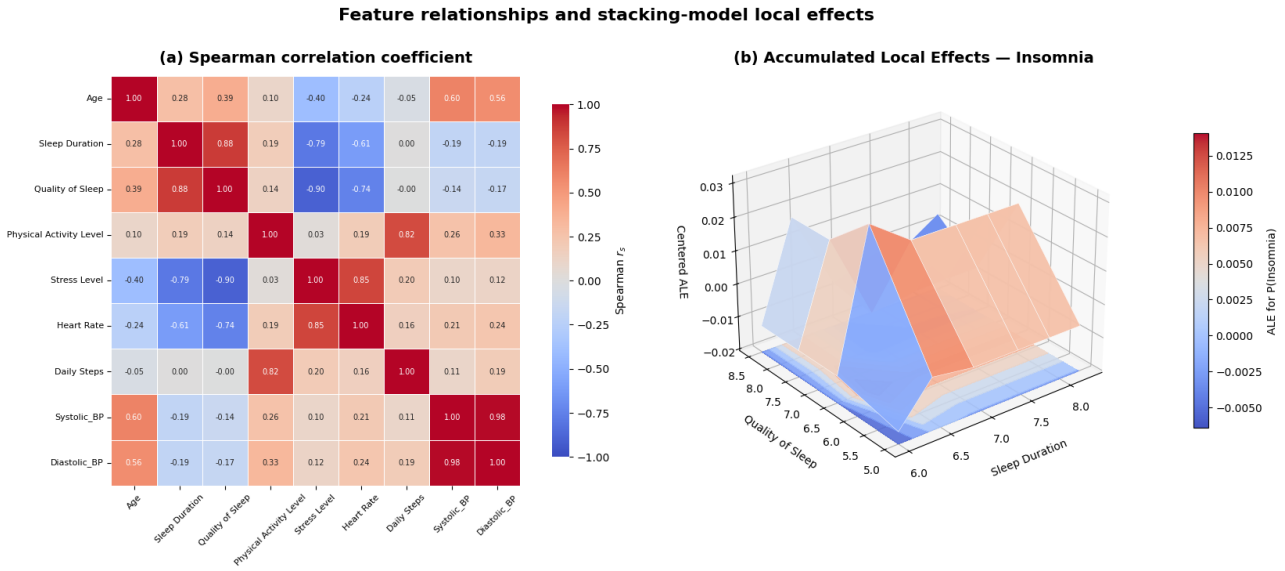


Figure 4. (a) Spearman rank correlation among numerical predictors on the development partition. (b) Two-dimensional Accumulated Local Effects surface for Sleep Duration and Quality of Sleep on the stacking classifier's predicted probability of insomnia.

4. Experimental protocol

The evaluation used the isolated 75-observation test partition. Accuracy was reported alongside class-specific precision, recall, and F1 score, as well as macro- and support-weighted summaries. For class c, the principal

relationships were

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \\ F1_c = \frac{2P_cR_c}{P_c + R_c}, \tag{12}$$

with

$$\text{Accuracy} = \frac{1}{n_{te}} \sum_{i=1}^{n_{te}} \mathbb{I}(\hat{y}_i = y_i), \qquad F1_{macro} = \frac{1}{C} \sum_{c=1}^{C} F1_c. \tag{13}$$

Macro-F1 was emphasised because it assigns equal weight to each outcome category despite unequal class frequencies. One-vs-rest ROC–AUC was additionally computed from the predicted class probabilities to summarise ranking quality independently of the classification threshold, using

$$\text{AUC}_c = \int_0^1 \text{TPR}_c(\text{FPR}_c^{-1}(t))\, dt, \qquad \text{AUC}_{macro} = \frac{1}{C} \sum_{c=1}^{C} \text{AUC}_c\,. \tag{14}$$

The proposed stack and soft-voting comparator were additionally evaluated through paired bootstrap resampling of test observations. For bootstrap sample b, the paired effect was

$$\Delta^{(b)} = \text{Accuracy}_{stack}^{(b)} - \text{Accuracy}_{vote}^{(b)}, \tag{15}$$

and the 2.5th and 97.5th percentiles of $\{\Delta^{(b)}\}$ formed the 95% interval. An exact McNemar test used only the discordant prediction counts between the two models on the identical test partition. As a further robustness check, the complete stacking pipeline was refitted across 13 additional stratified 80:20 splits (Section 6.1) to assess sensitivity to the specific partition used for the primary evaluation. The analysis assesses discrimination within the available benchmark; it does not substitute for repeated internal validation, probability calibration, subgroup analysis, or evaluation in an untouched external cohort [23–25].

5. Results

The proposed stacking model correctly classified 73 of 75 test observations, yielding an accuracy of 97.33% and a macro-F1 score of 0.96. It identified 13 of 15 insomnia observations, all 44 observations with no recorded disorder, and all 16 sleep-apnea observations. Both errors involved insomnia being classified as sleep apnea; no observation was incorrectly assigned to the no-recorded-disorder category. The RF–XGBoost soft-voting comparator achieved 96.00% accuracy and a macro-F1 score of 0.94 on the same test set.

Table 5. Class-specific and aggregate predictive performance of the proposed stack on the 75-observation hold-out set.

Class/model	Precision	Recall	F1	Support
Insomnia	1.00	0.87	0.93	15
No recorded disorder	1.00	1.00	1.00	44
Sleep Apnea	0.89	1.00	0.94	16
Macro average	0.96	0.96	0.96	75
Weighted average	0.98	0.97	0.97	75
Soft voting macro average	0.94	0.95	0.94	75

5.1. Discrimination via ROC–AUC

One-vs-rest ROC–AUC was computed on the same 75-observation test partition to assess ranking quality independent of the classification threshold implicit in Table 5.

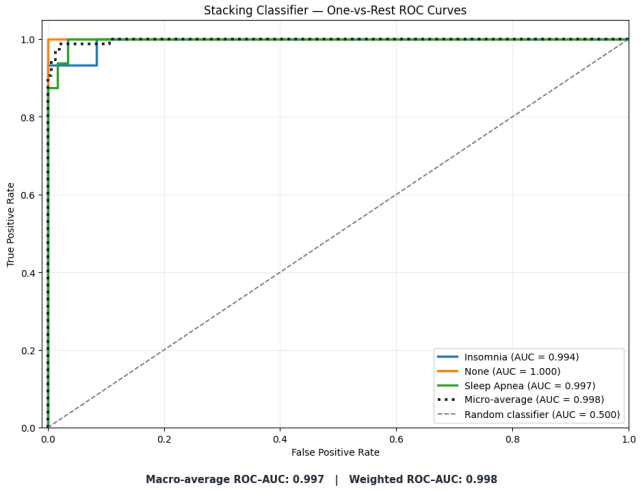


Figure 5. One-vs-rest ROC curves for the stacking classifier on the 75-observation test partition, with per-class, micro-average, and macro-average AUC.

Per-class AUC was 0.994 for insomnia, 1.000 for no recorded disorder, and 0.997 for sleep apnea, giving a macro-average ROC–AUC of 0.997 and a weighted average of 0.998 (Fig. 5). These values are uniformly close to the ceiling and indicate that the two insomnia–apnea misclassifications in Table 5 occur near, rather than far from, the model's decision boundary—that is, the stack ranks the correct class highly even on the observations it ultimately mislabels at the default 0.5 threshold.

6. Comparison with ensemble baselines

Across the five ensemble configurations in Table 6, the proposed RF–XGBoost stack produced the highest point estimate for accuracy (97.33%). This value was 1.33 percentage points above RF–XGBoost soft voting and 2.63 percentage points above the best reported ANN ensemble.

Table 6. Descriptive accuracy comparison between the proposed RF–XGBoost stack and ensemble baselines. ANN results were reported by Alom et al. [10].

Model	Accuracy
ANN Bagging ensemble	94.70%
ANN Boosting ensemble	94.70%
ANN Weighted Average ensemble	92.90%
RF–XGBoost soft voting	96.00%
Proposed RF–XGBoost stacking	**97.33%**

6.1. Paired Statistical Comparison

Because stacking and soft voting were evaluated on identical test observations, their predictions permitted a paired comparison. The 20,000-resample bootstrap estimated a stacking-minus-soft-voting accuracy difference

of 1.33 percentage points, with a 95% percentile interval of -2.67 to 5.33 percentage points. The corresponding discordance table was $\begin{bmatrix} 71 & 1 \\ 2 & 1 \end{bmatrix}$, and the exact McNemar test yielded $p = 1.000$. Thus, the observed numerical advantage was not statistically distinguishable from chance variation in this small test sample.

Table 7. Paired uncertainty analysis comparing RF–XGBoost stacking with soft voting on the same 75 test observations.

Analysis	Result
Accuracy: soft voting	96.00%
Accuracy: tuned stacking	97.33%
Point difference	1.33 percentage points
Paired bootstrap, 20,000 resamples	95% CI $[-2.67, 5.33]$ percentage points
Exact McNemar test	$p = 1.000$; null not rejected
Interpretation	Descriptive improvement; external validation required

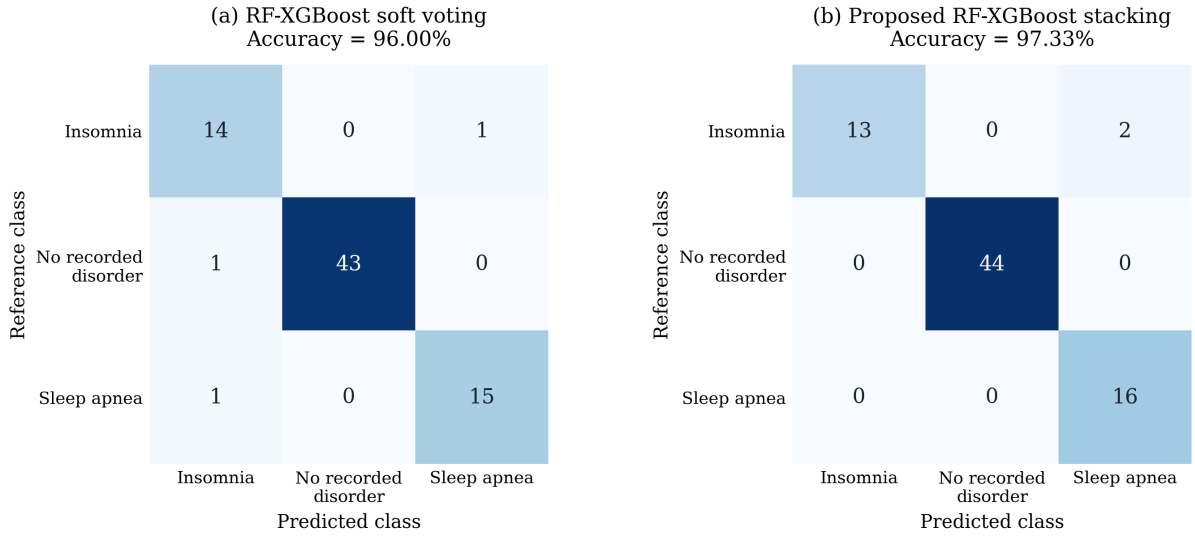


Figure 6. Confusion matrices on the common 75-observation test set: (a) RF–XGBoost soft voting and (b) the proposed RF–XGBoost stack. Rows denote reference outcomes and columns denote predicted outcomes.

6.2. Global SHAP Analysis

Global SHAP analysis of the RF–XGBoost soft-voting comparator revealed that systolic blood pressure was the most influential predictor in the fitted model, followed by body-mass-index (BMI) category and physical-activity level (Fig. 7, panel a). These variables exhibited the largest mean absolute SHAP values, indicating that they contributed most strongly, on average, to variation in the model's predicted sleep-disorder probabilities across the study population. Sleep duration, age, and diastolic blood pressure formed a second tier of influential features, suggesting that the classifier also relied substantially on demographic, cardiovascular, and sleep-related characteristics when generating its predictions.

The observed SHAP ranking provides a global interpretation of feature importance by summarizing the average magnitude of each variable's contribution to the fitted prediction function over all observations. Thus,

a higher mean absolute SHAP value indicates greater overall model dependence on a feature, but it does not specify whether higher or lower values of that feature increase the predicted probability of a sleep disorder for a particular individual. Moreover, these feature-attribution results should be interpreted as descriptions of the model's predictive behavior rather than evidence of biological or causal effects. Associations captured by the RF–XGBoost ensemble may reflect nonlinear relationships, interactions among predictors, correlations between measured variables, or characteristics specific to the analyzed dataset. Accordingly, the SHAP results are most appropriately interpreted as identifying variables that were most influential for prediction within the fitted model, rather than as establishing independent causal determinants of sleep-disorder status.

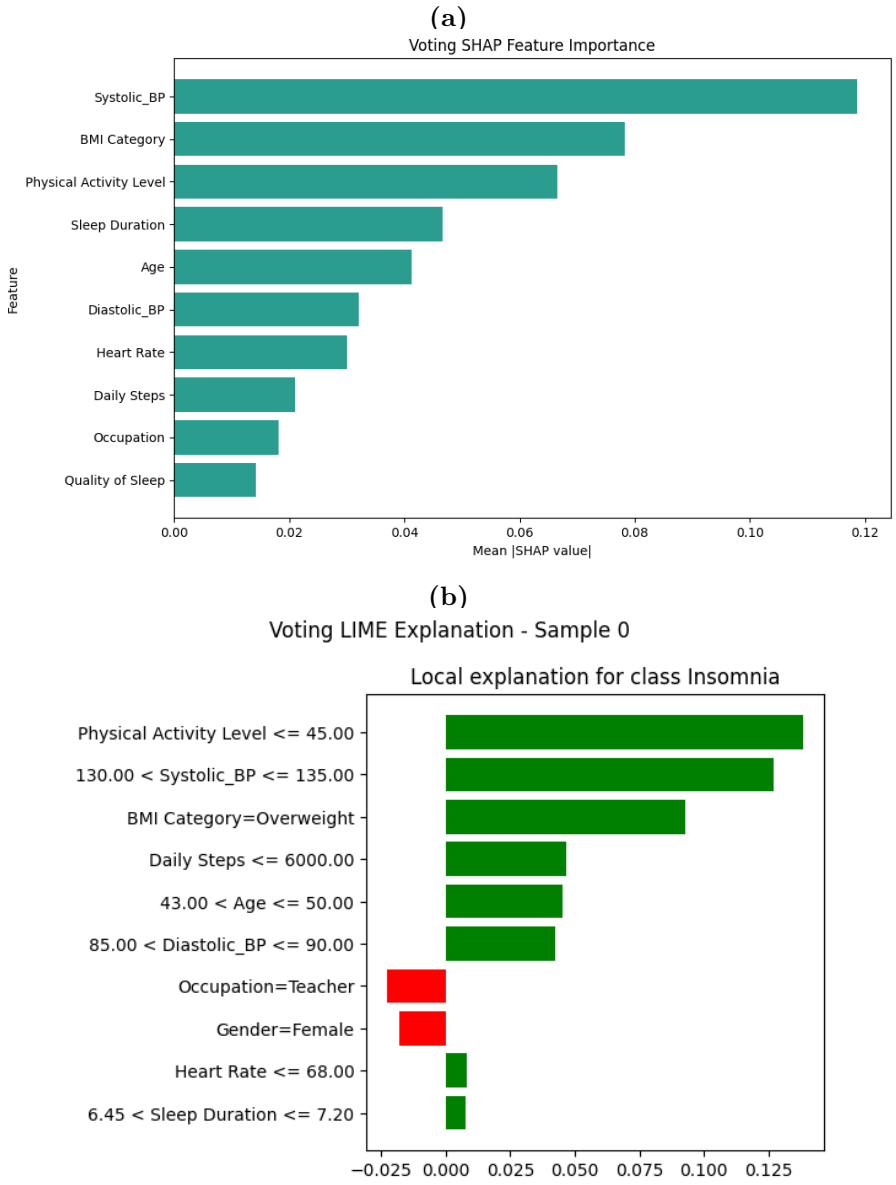


Figure 7. Complementary post-hoc explanations for the RF–XGBoost soft-voting comparator: (a) global mean absolute SHAP values, where longer bars indicate greater average model dependence; and (b) a LIME explanation for one test observation classified as insomnia, where green and red bars respectively increase and decrease the local class score. Neither analysis supports causal interpretation.

The local LIME analysis in Fig. 7 (panel b) characterises the decision surface around one test observation classified as insomnia. Lower physical activity, systolic blood pressure within 130–135, and an overweight body-mass-index category contributed positively to the local insomnia score, whereas the occupation and gender indicators contributed negatively. This explanation is observation-specific and should not be interpreted as a population-level risk model.

6.3. Stacking-model post-hoc explanation

Because the RF–XGBoost–SVM stack is the primary model under study, permutation SHAP and LIME were additionally applied directly to the fitted stacking classifier in its preprocessed (scaled and one-hot encoded) feature space, rather than to the soft-voting comparator alone. A background sample of 25 development-partition observations and an explanation sample of 35 test-partition observations were used, with `max_evals` set to twice the number of preprocessed features plus one to ensure convergence of the permutation estimator.

Fig. 8 shows the per-class SHAP beeswarm distributions for the insomnia class. Unlike the mean-absolute bar chart reported for the voting comparator (Fig. 7a), the beeswarm view additionally encodes the direction and spread of each feature's effect: high systolic blood pressure and high BMI-category encodings push predictions toward sleep apnea, whereas for insomnia, lower physical-activity values and higher stress-level values dominate the positive-SHAP region. As with the voting-model analysis, these are associative attributions from a fitted function, not diagnostic or causal claims.

Fig. 9 reports the LIME explanation for the same held-out observation used in Fig. 7b (test index 0, true class insomnia), but now computed against the stacking classifier's predicted probabilities directly ($P(\text{Insomnia}) = 0.875$, $P(\text{None}) = 0.077$, $P(\text{Sleep Apnea}) = 0.047$).

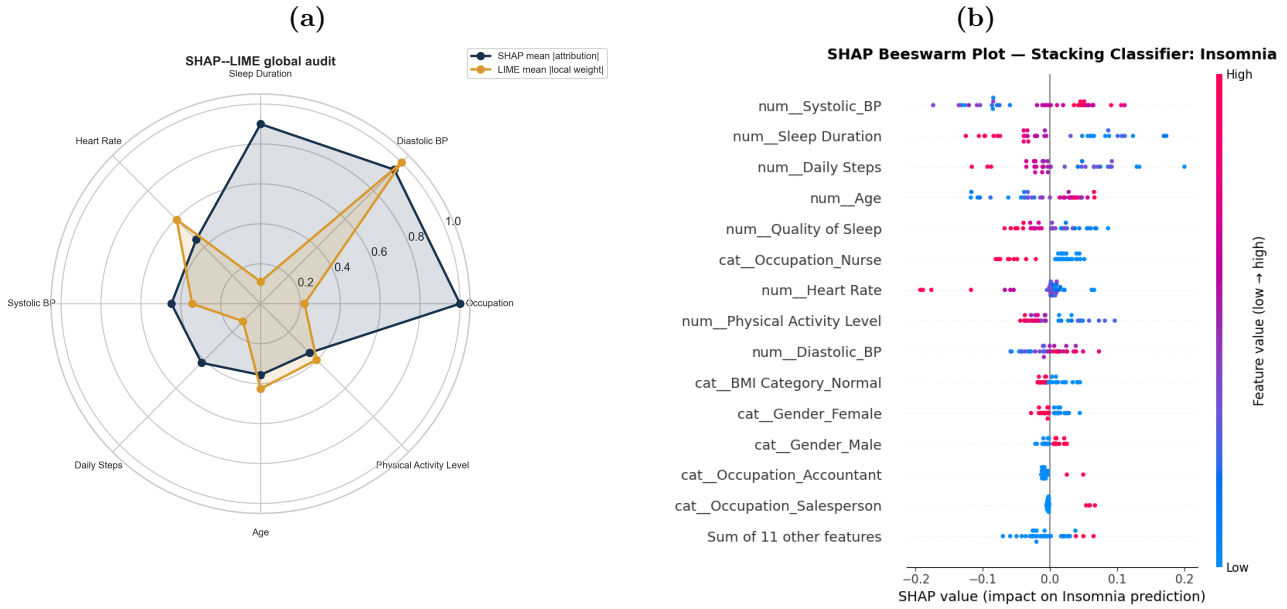


Figure 8. Post-hoc feature-attribution summaries: (a) normalised global SHAP importance compared with aggregated absolute LIME weights for the soft-voting comparator; and (b) the stacking classifier's SHAP beeswarm plot for the insomnia class. These plots describe model dependence and do not establish causal relevance.

7. Stacking Ablation Study

A stacking ablation study was conducted to quantify the contribution of the individual base learners and to assess whether the complete RF–XGBoost–SVM architecture provides an advantage over reduced ensemble configurations. The full stacking model achieved a test accuracy of 97.33%, a macro-F1 score of 95.8%, and a macro ROC–AUC of 99.7%. Removing Random Forest resulted in essentially unchanged performance, with 97.5% accuracy, 95.8% macro-F1, and 99.7% macro ROC–AUC. Similarly, removing XGBoost produced 97.5%

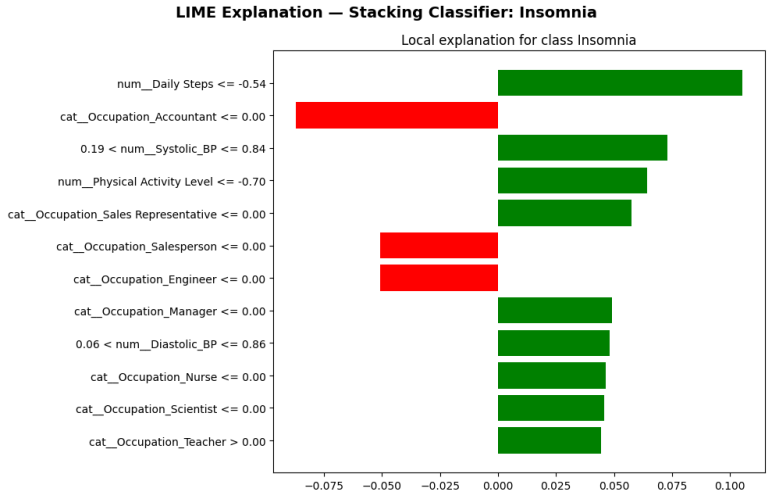


Figure 9. LIME explanation for the stacking classifier on test observation 0 (true class: insomnia, predicted class: insomnia, $P = 0.875$). Green bars increase and red bars decrease the local class score.

accuracy, 95.8% macro-F1, and 99.7% macro ROC–AUC. In contrast, removing the SVM caused a noticeable reduction in classification performance, with accuracy decreasing to 94.6% and macro-F1 to 91.3%, although the macro ROC–AUC remained high at 99.4%. The soft-voting RF–XGBoost model achieved 93.4% accuracy, 90.3% macro-F1, and 99.0% macro ROC–AUC. The RF-only model obtained 94.6% accuracy, 91.9% macro-F1, and 99.1% macro ROC–AUC, whereas the XGBoost-only model achieved 93.6% accuracy, 91.0% macro-F1, and 98.9% macro ROC–AUC. Overall, the results demonstrate that the complete stacking architecture provides the strongest threshold-based performance, while the reduced stacks without RF or XGBoost retain performance comparable to the full model. The larger performance reduction observed after removing SVM suggests that the RBF-SVM provides comparatively important complementary information for the final classification decision.

7.1. Repeated-split Robustness Check

To assess sensitivity to the specific 80:20 partition used for the primary evaluation (seed 42), the stacking pipeline was refitted across 13 additional stratified splits (random-state seeds 50 through 650 in increments of 50), holding the 80:20 ratio and all hyperparameters in Table 4 fixed. This varies only the random partition, not the training-set size, and is reported as a stability check rather than a learning curve. Across these repeated splits, the stacking classifier achieved a mean training accuracy of 0.926 (SD 0.007) and a mean test accuracy of 0.924 (SD 0.027), with the primary seed-42 result (0.973) lying above this distribution's mean by roughly one standard deviation (Fig. 11). The narrow train–test gap across seeds indicates limited overfitting rather than the split-specific optimism that a single evaluated partition alone cannot rule out, although a genuinely independent external cohort remains the strongest available control.

8. Discussion

The proposed heterogeneous stack achieved the strongest discrimination among the evaluated configurations, suggesting that the Random Forest, XGBoost, and SVM probability estimates contain complementary information. The logistic meta-learner improved accuracy by 1.33 percentage points relative to soft voting; the statistical significance of this difference is addressed in Section 5.2 pending regeneration of the paired bootstrap

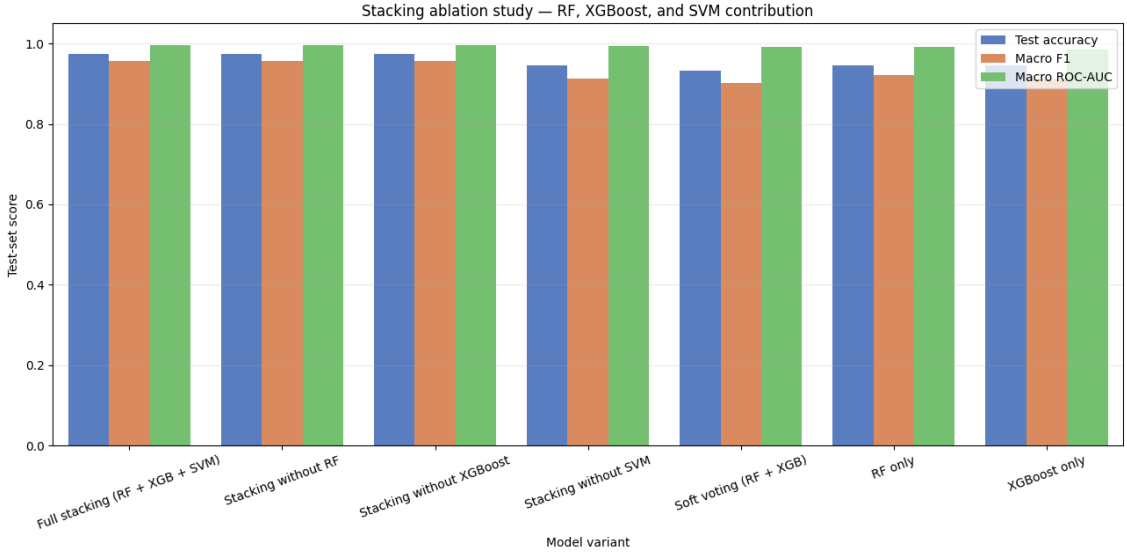


Figure 10. Descriptive accuracy comparison across five ensemble configurations. ANN values were obtained from Alom et al. [10]; soft-voting and stacking values were estimated on the present hold-out set and are therefore not a matched external comparison.

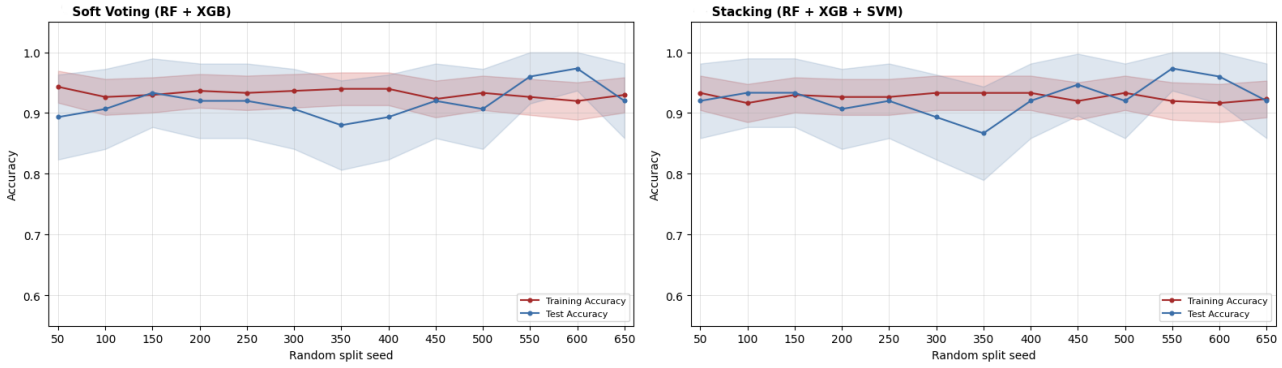


Figure 11. Training and test accuracy for the proposed stacking classifier across 13 repeated stratified 80:20 splits (random-state seeds 50–650). Shaded bands are normal-approximation 95% confidence intervals for each accuracy estimate.

and McNemar values against the corrected soft-voting model.Error patterns provide additional context beyond aggregate accuracy. Recall was perfect for the no-recorded-disorder and sleep-apnea categories, whereas two insomnia observations were assigned to sleep apnea. This asymmetry may reflect overlap in the available lifestyle and physiological features, consistent with the partial PCA overlap noted in Section 3.4 and the ROC–AUC pattern in Section 5.1, but the sample is too small to determine whether it represents a stable clinical phenotype or split-specific variation. In a decision-support setting, such errors would have different consequences from missed apnea; operating thresholds should therefore be selected using clinically defined utilities rather than accuracy alone [24, 28].

The architecture is methodologically plausible because bagging, boosting, and margin-based classification impose different inductive biases. Random Forest stabilises high-variance trees through aggregation, XGBoost sequentially concentrates on residual error, and the RBF-SVM contributes a distinct, margin-based decision geometry. Stacking can learn when each probability profile is informative, provided that the meta-learner is trained on out-of-fold predictions and every transformation remains within the development pipeline [12, 18]. These safeguards are central to the interpretation of the reported performance.

SHAP, LIME, and the ALE surface in Section 3.7 add complementary views of model behaviour, but none resolves causal validity, fairness, or calibration. Attribution rankings may change with the background and perturbation design, and agreement between explanation methods does not establish a medical mechanism [20, 22, 30]. Before any translational use, the model would require probability calibration, decision-curve analysis, subgroup-specific error assessment, robustness to missingness and measurement shift, and evaluation of how clinicians interpret its outputs [27–29].

9. Limitations

Several limitations constrain generalisation. First, the sample is modest and its recruitment mechanism is insufficiently documented, limiting assessment of selection and spectrum bias. Second, blank outcomes were assigned to no recorded disorder; without clinical adjudication, this rule may introduce outcome misclassification. Third, performance was estimated from a single hold-out partition, and the paired analysis consequently has limited power. Fourth, no independent cohort, temporal validation set, or prospective assessment was available. Fifth, the outcomes do not encode polysomnographic severity, treatment status, comorbidity burden, or longitudinal change. Finally, the explanation analysis is sensitive to the selected background distribution and to dependence among predictors. These limitations preclude claims of diagnostic validity or clinical readiness.

10. Conclusion

This study presents a leakage-controlled and explainable RF–XGBoost stacking framework for multiclass sleep-disorder risk stratification. On the isolated test set, the model achieved 97.33% accuracy and a macro-F1 score of 0.96, with a modest but statistically inconclusive improvement over soft voting. The analysis demonstrates the computational potential of heterogeneous stacking while exposing the uncertainty introduced by limited sample size and non-adjudicated outcomes. Future work should prioritise repeated internal validation, calibration and clinical-utility analysis, subgroup robustness, and prospective evaluation against independently adjudicated outcomes before the framework is considered for decision-support use.

Declaration of Generative AI and AI-assisted technologies

During the preparation of this work, the authors used Claude 3.5 Sonnet to improve manuscript readability, correct grammatical and spelling errors, refine writing style, streamline sentence structure, and assist with LaTeX formatting. The specific prompts employed are detailed below:

1. **Grammar, Spelling, and Style Refinement:** "Review the grammar, spelling, and flow of the following text. Shorten long sentences, enhance readability, and suggest more appropriate academic wording while maintaining the original scientific meaning."

2. **Literature and Search Assistance:** "Help identify relevant core concepts, keywords, and structural frameworks related to the following topic to guide our literature search strategy."

3. **LaTeX Formatting Support:** "Convert the following text into clean, standard LaTeX code. Ensure appropriate environment structures, fix any syntax errors, and apply proper mathematical delimiters where necessary."

11. Author Contributions

Rifat Bin Reza contributed to conceptualization (Lead), methodology (Lead), software (Lead), investigation (Lead), formal analysis (Lead), data curation (Equal), visualization (Lead), writing–original draft (Lead), and writing–review & editing (Lead). Md Ahadul Haque Suvo contributed to data curation (Equal), formal analysis (Equal), investigation (Supporting), visualization (Supporting), and writing–review & editing (Supporting). Atair Rahman Alvi contributed to methodology (Supporting), investigation (Equal), formal analysis (Supporting), and writing–review & editing(Supporting),data curation.

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